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Tran Minh Duc
SNR International Investment
Joint Stock Company,
Viet Nam

Trademark management model in the digital environment: Singapore's experience and its applicability in Vietnam

Tran Minh Duc

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Abstract

Currently, e-commerce is becoming a pillar of the modern economy, creating numerous opportunities while also posing major challenges in trademark protection and management. The value of a trademark is no longer confined to traditional signs but extends into the digital space, encompassing elements such as domain names, images, logos, and online reputation. Protecting and managing trademarks in this environment requires a close integration of technology, law, and corporate governance strategies. Experience from Singapore a pioneer in intellectual property management shows that an effective model should be built upon a transparent legal framework, comprehensive digitalization of management processes, and enhanced collaboration between the public and private sectors. Vietnam has made certain progress in improving its legal framework and strengthening the supervision of e-commerce activities; however, it still faces significant difficulties in detecting and addressing online trademark infringements. Therefore, learning from Singapore's experience is of practical importance for Vietnam in developing a modern, proactive, and adaptive trademark management model that aligns with the digital economy era, thereby enhancing competitiveness and protecting national brand value in the context of globalization.

Keywords: Trademark management, intellectual property, trademark infringement, digital trademarks, online platform liability

1. Introduction

In the era of the digital economy, e-commerce has become a key driver of global economic growth. Corporations such as Amazon, Alibaba, and eBay are leading an industry worth trillions of US dollars, fundamentally transforming business and consumer models on an international scale. In Vietnam, the e-commerce market surpassed USD 20 billion in 2023, with an annual growth rate of approximately 20%, demonstrating the increasingly significant role of the digital economy in national development. In this context, trademarks play a particularly important role, serving both as identifiers and as intellectual property assets with substantial commercial value, enabling enterprises to affirm their credibility and competitive capacity. However, the borderless and intangible nature of the digital environment poses significant challenges for trademark administration and protection. Infringement activities such as counterfeiting, unauthorized use, and trademark misappropriation on e-commerce platforms and social media have become increasingly prevalent, causing serious harm to businesses and undermining consumer trust. Vietnam's current legal framework still largely relies on traditional regulatory models, which have yet to fully accommodate the unique characteristics of the digital space. Given this situation, researching trademark management models in the digital environment is necessary to strengthen legal effectiveness and promote fair and healthy commerce. Singapore, a pioneering nation in the digital economy and intellectual property governance, has developed a modern management model based on digitalization, transparency, and international cooperation. This article analyzes Singapore's experience in digital trademark management, thereby drawing lessons and proposing recommendations for Vietnam to improve its legal mechanisms, ensuring a safe, transparent, and sustainable e-commerce environment.

2. Overview of trademark management models in the digital environment

2.1 Theoretical Basis for Trademark Management in the Digital Environment

In the context of the digital economy, the concept of trademarks has gone far beyond the traditional scope of physical identifiers affixed to goods or packaging.

Correspondence
Tran Minh Duc
SNR International Investment
Joint Stock Company,
Viet Nam

Today, trademarks encompass digital assets such as domain names, online logos, brand imagery and content on social media, as well as reputational value built in the digital environment. This reflects the global shift of commerce from physical space to cyberspace, where brand value lies not only in tangible products but also in data, online experiences, and consumer interactions. Accordingly, trademark management in the digital environment becomes a complex challenge, requiring coordination among legal frameworks, technological solutions, and brand governance strategies. Beyond the traditional trademark registration process, businesses must proactively monitor, detect, and address infringement activities occurring online, ranging from cybersquatting and logo duplication to unauthorized use of brand images on e-commerce platforms or social networks. According to Thomson Reuters (2022), trademark protection in the digital era “requires going beyond the scope of traditional registration”, as the online environment creates sophisticated infringement methods that spread far more rapidly than in physical markets ^[1].

In response to the increasingly complex demands of brand protection in cyberspace, many countries have proactively adopted advanced technological solutions such as Artificial Intelligence (AI), big data analytics, image-recognition algorithms, and online trademark databases to support monitoring and detection of infringement. These tools enable regulatory authorities and enterprises to track, verify, and address violations rapidly, accurately, and cost-effectively. Alongside the deployment of technological applications, a multi-stakeholder governance mechanism has become essential. This requires close coordination among intellectual property authorities, digital platforms, businesses, and intermediary organizations to build a comprehensive trademark protection ecosystem. Such collaboration not only enhances the effectiveness of legal enforcement but also contributes to establishing a transparent, fair, and secure e-commerce environment, thereby ensuring the sustainable development of trademarks in the era of global digitalization.

Trademark infringement in the online environment has become increasingly sophisticated and difficult to control, making early monitoring and prevention at the product-upload stage a strategic approach in digital trademark management. Rather than merely addressing violations after they occur, businesses and regulatory authorities are now shifting toward a proactive management model, emphasizing early detection and risk prevention from the very first step of the online transaction process.

Automated scanning technologies are being widely deployed to identify infringing content, detect images, descriptions, or keywords that indicate potential trademark counterfeiting before products are made publicly available on e-commerce platforms. In addition, registering and protecting domain names associated with the brand is regarded as a crucial defensive measure to prevent phishing attacks and domain hijacking. If left uncontrolled, such acts can seriously damage brand reputation and erode consumer trust. Accordingly, modern digital trademark management strategies recommend combining technological tools, online monitoring systems, and rapid-response mechanisms as indispensable components. The objective is not only to detect and address violations but also to build a proactive, intelligent, and sustainable governance system, ensuring comprehensive trademark protection in the digital space,

where the boundary between legal protection and technological management is increasingly blurred ^[2].

In addition to technical and technological measures, the theoretical foundation of trademark management in the digital environment also emphasizes the importance of internal legal instruments, particularly contractual mechanisms and brand-governance policies within enterprises. Establishing clear legal agreements, such as franchise contracts, trademark licensing agreements, confidentiality agreements, and intellectual property management clauses in digital-technology contracts, plays a crucial role in controlling and protecting trademark assets. In the digital environment, trademarks are not only exploited directly by their owners but also used, distributed, or reproduced through partners, agents, distributors, media units, and intermediary platforms. Therefore, clearly defining the scope of use rights, confidentiality obligations, responsibilities for handling infringements, and dispute-resolution mechanisms in contractual arrangements serves as a legal shield to mitigate risks of intellectual property asset leakage. Particularly in the context of increasingly widespread digital collaboration from technology transfer and online marketing to brand-data governance internal corporate policies on information management, system access, and brand-image protection must be aligned with intellectual property regulations. This approach not only protects trademarks against external infringements but also strengthens internal compliance and self-defense capabilities, forming a foundation for a sustainable and adaptable trademark-management model in the digital era ^[3].

Within the modern legal framework, developing a trademark-management model in the digital environment requires special emphasis on regulations governing the handling of trademark infringements on online platforms, in order to ensure the effectiveness and efficiency of industrial property protection mechanisms. The rise of e-commerce platforms, social networks, and digital services has led to the emergence of new forms of infringement such as copying, counterfeiting, and unauthorized use of trademarks, requiring the legal system to adapt flexibly. Accordingly, the law must clearly define the responsibilities of intermediaries, including platform operators, network service providers and data-hosting entities in monitoring, detecting, and coordinating enforcement against infringing activities. In addition, establishing a rapid, transparent, and verifiable mechanism for removing infringing content is essential to safeguard legitimate trademark-holder rights while maintaining a safe and equitable e-commerce environment. Such a comprehensive legal model not only enhances the enforcement of intellectual property rights but also strengthens business and consumer confidence in the digital marketplace.

2.2 Trademark management model in the digital environment under Singaporean Law

Singapore is regarded as one of the leading nations in the region in terms of innovation and a modern Intellectual Property (IP) legal framework, providing a solid foundation for the development of a knowledge-based economy. To strengthen its position as Asia's intellectual property hub, the Singaporean Government introduced the Singapore IP Strategy 2030 (SIPS 2030), which sets the direction for developing IP laws, infrastructure, and services toward

digitalization, transparency, and business-friendliness. Within this strategic framework, two significant legislative instruments, the Intellectual Property (Amendment) Act 2022 and the Copyright Act 2021 were enacted to update and adapt legal provisions to the unique characteristics of digital technologies, electronic transactions, and online business environments.

With respect to trademark management, Singapore continues to adhere to the “first-to-file” principle, under which the right to registration belongs to the first applicant, thereby ensuring clarity and stability in the registration system. However, in order to strengthen the protection of legitimate rights and prevent trademark squatting, the Intellectual Property (Amendment) Act 2022 introduced the concept of “bad faith,” allowing the competent authority to invalidate trademark registrations obtained with dishonest or improper intention. This represents an important step forward in enhancing the protection mechanism and promoting fairness in market competition. In addition to legislative reforms, the Intellectual Property Office of Singapore (IPOS) has accelerated digital transformation in trademark administration, exemplified by the IPOS Go mobile application, the world’s first tool integrating the “Brand Search” function, enabling users to simultaneously search similar business names, trademarks, domain names, and social media accounts with a single scan. The combination of institutional innovation and technological advancement demonstrates Singapore’s clear orientation toward building an integrated trademark-management model within the digital ecosystem, ensuring both legal effectiveness and fostering innovation in the knowledge-based economy ^[4]. In terms of law enforcement in the digital environment, Singapore is regarded as one of the jurisdictions with a clear and consistent legal framework for determining the responsibilities of digital intermediaries. Under the current legal provisions, network service providers including internet infrastructure operators, telecommunications providers, social networking platforms, and e-commerce platforms are granted immunity from civil and criminal liability for intellectual property infringements committed by users on their platforms. This safe-harbour mechanism is intended to encourage the development of digital infrastructure and ensure a balanced approach between the rights of IP holders and the obligations of intermediaries, similar to the legal models adopted in the European Union and the United States. However, the scope of immunity in Singapore does not extend to copyright infringements, as this area is governed by specific statutory provisions and dedicated enforcement mechanisms. Notably, according to discussions at the 2022 AIPPI Congress, e-commerce platforms in Singapore are considered a form of Network Service Provider (NSP) and may therefore benefit from liability exemption for trademark, patent, or industrial design infringements committed by sellers on their platforms, provided that they fulfil their obligations relating to monitoring, takedown procedures, and cooperation with competent authorities upon request. This approach demonstrates the flexibility and balance in Singapore’s legal regime, simultaneously fostering innovation while maintaining an effective and context-appropriate mechanism for protecting intellectual property rights in the digital environment.

In addition to the legal framework on liability exemptions for network service providers, judicial practice in Singapore

has further clarified the limits and legal obligations of e-commerce platforms in handling trademark infringements. According to the Singapore courts, e-commerce platforms act merely as intermediaries, similar to a “digital marketplace”, and therefore are not directly liable for trademark infringements committed by sellers, provided that they promptly remove infringing content upon receiving a valid notice from the rights holder or a competent authority. However, where a platform has evidence of, or is clearly aware of, infringing activity yet intentionally fails to take action, the court may determine that it has participated in the wrongdoing under the applicable common-law principles and hold it jointly liable with the infringing party. This position reflects the flexible approach of Singapore’s judicial system, which protects the legitimate interests of trademark owners while avoiding excessive liability for platforms that merely provide technical infrastructure. It should be noted that, unlike in the area of copyright where Singapore has a specific statutory safe-harbour regime, there is no equivalent mechanism governing trademark infringements. Instead, Singapore applies a notice-and-takedown model as a flexible mechanism that encourages proactive cooperation between trademark owners and e-commerce platforms in detecting, addressing, and preventing infringements. This mechanism is regarded as an effective compromise between safeguarding intellectual property rights and maintaining a digital environment that supports openness, competition, and innovation ^[5].

In addition to strengthening the legal framework, Singapore has continuously reformed its administrative procedures for trademarks to adapt to the trend of digitalization and maximize convenience for businesses. Singapore’s intellectual property legislation has introduced a partial-acceptance mechanism for trademark applications, allowing portions of an application that satisfy protection requirements to be approved first, while the remaining components may continue to be reviewed or amended. This mechanism shortens registration time, reduces costs, and increases flexibility for applicants. Furthermore, the new regulations expand the right to seek third-party comments prior to the issuance of a trademark certificate, thereby enhancing transparency, preventing disputes, and improving examination quality. Notably, the Intellectual Property Office of Singapore (IPOS) is internationally recognized as one of the most innovative and dynamic IP authorities in the world, with a vision of building a “smart and fully digitalized intellectual property ecosystem.” IPOS continuously upgrades its online filing systems, provides direct and virtual IP advisory services for businesses, and develops platforms to support smart search, monitoring, and portfolio management for trademarks. These reforms not only simplify administrative procedures but also improve the efficiency of trademark protection, enforcement, and commercialization in the digital environment, reaffirming Singapore’s pioneering position in intellectual property governance in the Asia-Pacific region.

2.3 Trademark management model in the digital environment under Vietnamese Law

Vietnamese trademark law is primarily governed by the Law on Intellectual Property 2005, as amended in 2009, 2019, and 2022. Under this law, a trademark is defined as a sign used to distinguish goods and services, with protection granted mainly through the registration process. Although

the law has not yet introduced a specific concept of a “digital trademark,” it has been updated with several new provisions to accommodate technological developments. Notably, the 2022 amendments to the Law on Intellectual Property introduced the concept of “bad faith” as a legal basis for invalidating trademark registrations obtained with dishonest intent, thereby addressing shortcomings in the “first-to-file” system. This provision strengthens the legal tools available to Vietnamese enterprises to oppose or invalidate trademark applications filed by competitors seeking to unlawfully appropriate brand assets. In addition, the amended law expands the scope of protectable subject matter by allowing the registration of sound marks and clarifying the conditions for granting certification marks and collective marks. These provisions support brand management across diverse digital platforms (for instance, sound branding used in digital applications).

In terms of law enforcement in the digital environment, Vietnam has begun establishing a legal framework governing e-commerce and online advertising activities in connection with the protection of intellectual property rights, including trademarks. Under the Law on Electronic Transactions 2023 and its implementing decrees, e-commerce platforms are obligated to coordinate with intellectual property rights holders to review, detect, and remove infringing goods through a publicly disclosed notice-and-takedown procedure. However, in practice, the monitoring and handling of online trademark infringements still face significant challenges, primarily due to the difficulty in verifying the real identities of infringers. Most seller accounts use false or incomplete information, hindering investigation and enforcement activities. In addition, the recurring nature of infringements reduces long-term deterrence, as infringers can easily create new accounts after takedown measures and continue their activities. This situation indicates that Vietnam must further improve its legal framework while strengthening technical monitoring capabilities, data-sharing mechanisms, and multi-stakeholder cooperation between the State, businesses, and trademark owners, in order to enhance the effectiveness of intellectual property enforcement in the digital environment. According to a report by the Ministry of Science and Technology, between 2019 and the end of 2022, up to 60% of requests for intellectual property rights enforcement submitted to competent authorities were related to e-commerce activities^[6]. This figure demonstrates the rapid increase in trademark infringements in the digital environment, while also reflecting significant challenges in intellectual property enforcement in Vietnam. The strong growth of online platforms and digital transaction models has facilitated the widespread distribution of trademarked goods and services, but has also created space for sophisticated and difficult-to-control acts of counterfeiting, infringement, and unauthorized use of trademarks. This situation requires the legal system and regulatory authorities to innovate monitoring methods, strengthen technological applications, and cooperate with digital platforms to ensure the effectiveness of trademark protection in e-commerce a rapidly developing sector with substantial legal risks.

Recognizing the growing challenges posed by intellectual property infringements in the digital environment, Vietnamese regulatory authorities have been actively working to improve relevant policies and legal frameworks in order to enhance enforcement effectiveness. According to

the Intellectual Property Office of Vietnam (IP Viet Nam), the country must “accelerate the development of the legal framework and enforce the law effectively” to prevent intellectual property violations, particularly trademark infringements on e-commerce platforms. Sharing the same perspective, Deputy Director General Trần Lê Hồng emphasized the rapid development of e-commerce in the context of the global innovation-driven economy. He noted that intellectual property issues, especially trademarks play a crucial role and are among the key factors determining the success or failure of enterprises conducting business on e-commerce platforms^[7]. Accordingly, various measures are being implemented to ensure a balance between e-commerce development and the protection of intellectual property rights.

The key proposed solutions include: (i) strengthening specialized inspection and enforcement activities and imposing strict penalties on violators; (ii) applying information technology and artificial intelligence to monitoring, detecting infringements, and managing online trademark data; and (iii) raising public, business, and consumer awareness of the importance of trademark protection in the digital environment. These efforts demonstrate Vietnam’s direction toward building a proactive and effective trademark protection mechanism consistent with digital-transformation trends, thereby contributing to sustainable and equitable development of the national e-commerce market.

3. Lessons for Vietnam

Drawing on Singapore’s experience, several lessons can be identified to improve Vietnam’s digital trademark-management model:

Firstly, enhance the legal framework. Vietnam should continue updating and refining trademark legislation in a manner that strengthens the responsibilities of e-commerce platforms and digital intermediaries in preventing, detecting, and addressing infringements. It may consider adopting a “safe-harbour” mechanism similar to Singapore’s Copyright Act to incentivize platform compliance, while clearly defining notice-and-takedown procedures to ensure transparency and efficiency. The introduction of the “bad faith” doctrine in Vietnam’s 2022 IP Law is an important step forward; moving forward, Vietnam may consider adding mandatory notification requirements and joint-liability rules for platforms that “know or ought to know” of infringing activities, consistent with the “joint tortfeasor” concept under Singaporean law.

Secondly, strengthen enforcement and public-private cooperation. Vietnam should establish a transparent and expeditious enforcement mechanism based on Singapore’s experience with the notice-and-takedown model, involving coordinated action among the Intellectual Property Office of Vietnam, the Market Surveillance Authority, and e-commerce platforms. Platforms should maintain dedicated complaint-handling units and impose strict penalties on repeated offenders. In addition, Vietnam could promote public-private partnerships in trademark education and advisory services, similar to IP clinics in Singapore, to help enterprises better understand legal requirements, enhance self-protection capabilities, and mitigate risks.

Thirdly, adopt digital technologies in trademark management. Vietnam should accelerate the digitalization of trademark registration, search, and management systems,

drawing lessons from Singapore's IPOS Go and IP2SG platforms, which allow integrated searches across trademarks, domain names, and social media accounts. Vietnam should develop an integrated online portal for trademark filing and search, and deploy artificial intelligence and automated data-scanning tools on e-commerce platforms to detect infringements early, thereby reducing administrative burdens and improving monitoring efficiency.

Fourthly, overall strategic policy direction. Vietnam should formulate a national strategy on intellectual property and digital-era trademark protection similar to Singapore's IP Strategy 2030, aligned with digital-economy and innovation-development goals. This strategy should leverage international agreements (CPTPP, EVFTA, RCEP, etc.) to broaden protection scope, including images, sound marks, digital codes, and barcodes used in online environments. At the same time, the government should strengthen public awareness and legal education on trademarks, encourage Vietnamese enterprises to build strong brands and enhance compliance, thereby establishing a sustainable foundation for the national e-commerce ecosystem.

4. Conclusion

The study demonstrates that an effective trademark-management model in the digital environment requires a harmonious combination of a modern legal framework, technological application, and multi-stakeholder coordination. Singapore serves as a prime example of a pioneering approach, comprehensively digitalizing its intellectual property system, clearly defining the responsibilities of digital platforms, and promoting public-private cooperation to ensure transparency and efficiency in trademark protection. Meanwhile, Vietnam has made significant progress through the Law on Intellectual Property 2005, as amended in 2009, 2019, and 2022, along with implementing decrees. However, the country continues to face major challenges due to the rapidly increasing prevalence of online trademark infringements. Therefore, Vietnam must continue reforming its legal framework with greater flexibility, strengthening enforcement capacity, accelerating digital technology adoption, and enhancing international cooperation. Only by building a proactive, transparent, and digitally-adaptive trademark-management model can Vietnam effectively protect and promote its brand value, enabling Vietnamese brands to rise and compete within the global value chain.

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